Reliability Assessment of the JAXA H3 Fairing under Manufacturing Variability

Polynomial Chaos Expansion as a Stochastic FEM Surrogate

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Stochastic Finite Element Methods — Dr. Z. Zheng / Prof. U. Nackenhorst LUH SoSe 2026

Motivation: manufacturing variability in composite fairings

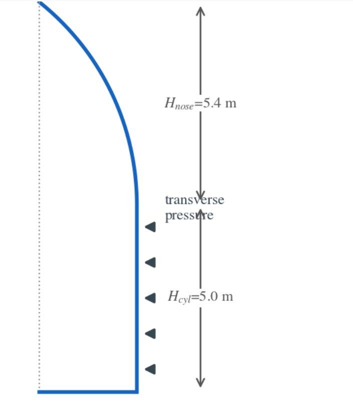
- The H3 fairing is a **CFRP / aluminum-honeycomb sandwich**.
- Material properties vary from manufacturing: fiber volume fraction, cure cycle, adhesive film thickness.
- This uncertainty propagates into **stress, damage, and deflection** — and hence reliability.

Goal: quantify how five uncertain material parameters propagate to the structural response, treating the Abaqus model as a black box.

Stochastic FEM

Propagate input uncertainty through a deterministic FE model to obtain response statistics and a failure probability.

The structure: H3 payload fairing — design & FE model



Geometry (CFRP / Al-honeycomb sandwich)

- Diameter 5.2 m; ogive nose $H_{\text{nose}} = 5.4$ m, cylinder $H_{\text{cyl}} = 5.0$ m
- CFRP face sheets 0.6–1.4 mm; Al-honeycomb core 8 mm

Material constants

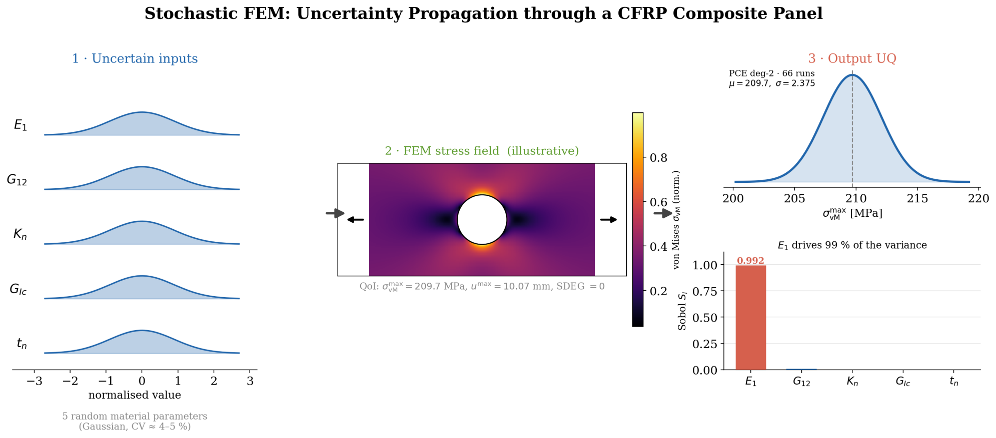
- CFRP skin (orthotropic LAMINA): $E_1 \approx 146$ GPa, $E_2 \approx 10$ GPa, $G_{12} \approx 5.2$ GPa, $\nu_{12} = 0.3$
- Al-honeycomb core: orthotropic — stiff out-of-plane (E_3), soft in-plane
- Adhesive — cohesive zone (BK, $\eta = 2.284$): $K_n = 10^6$ N/mm^3, $G_{Ic} = 0.3$ N/mm, $t_n = 50$ MPa

Boundary conditions & load

- Base ring **clamped**; skins–core tied via the cohesive interface
- Step 1: transverse **pressure**; Step 2: incremental load

Five constants ($E_1, G_{12}, K_n, G_{Ic}, t_n$) are treated as **uncertain** (next slides).

Approach at a glance



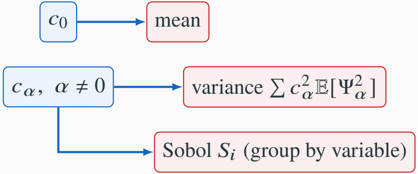
Non-intrusive: sample the five parameters, propagate each draw through the deterministic Abaqus model, recover the response distribution and Sobol indices.

Method: Polynomial Chaos Expansion

For a response $Y = \mathcal{M}(\xi)$ with random inputs $\xi \in \mathbb{R}^d$:

$$Y \approx \hat{Y} = \sum_{\alpha \in \mathcal{A}} c_\alpha \Psi_\alpha(\xi), \qquad P = \binom{d+p}{p}.$$

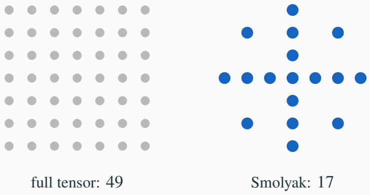
- Ψ_α: **Hermite** polynomials — orthogonal w.r.t. the Gaussian measure.
- The basis is *fixed by the input measure*, not chosen.
- It is the **orthogonality**, not the polynomial form, that pays.



No extra model runs — all three come from the coefficients.

Method: Smolyak sparse quadrature

Full tensor-product quadrature needs n_q^d points — prohibitive at $d = 5$. Smolyak combines low-order 1-D rules, keeping exactness on the **total-degree** space only.



(2-D illustration — the same idea at $d = 5$.)

p	Full	Sparse	Reduction
2	3^5=243	**66**	3.7×
3	4^5=1024	**286**	3.6×

Five uncertain input parameters

Parameter	Description	Nominal	CoV
E_1	fiber-direction modulus	146,000 MPa	**5%**
G_{12}	in-plane shear modulus	5,200 MPa	10%
K_n	cohesive normal stiffness	10^6 N/mm^3	15%
G_{Ic}	Mode-I fracture energy	0.3 N/mm	20%
t_n	cohesive strength	50 MPa	15%

Independent **truncated-normal** variables on $[0.5\mu,\ 1.5\mu]$ — stiffnesses and fracture energies are strictly positive, and flight hardware is screened against inspection limits.

In σ-units the bound is $0.5/\mathrm{CoV}$, so the truncation is **practically inactive** for every input — it guarantees positivity without reshaping the distribution.

CoV	bound	mass cut
5 % (E_1)	$\pm 10\sigma$	~ 0
15 %	$\pm 3.3\sigma$	0.09 %
20 % (G_{Ic})	$\pm 2.5\sigma$	1.2 %

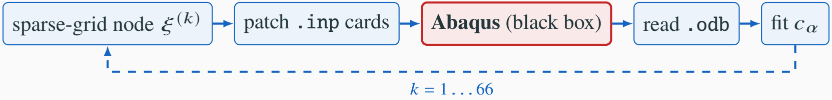
Finite-element model & quantities of interest

Abaqus/Standard panel model:

- CFRP skin: orthotropic LAMINA elastic.
- Al-honeycomb core: linear elastic.
- Adhesive: **cohesive zone model** (BK mixed-mode, $\eta = 2.284$).
- Two-step static load; QoIs from last frame via `odbAccess`.

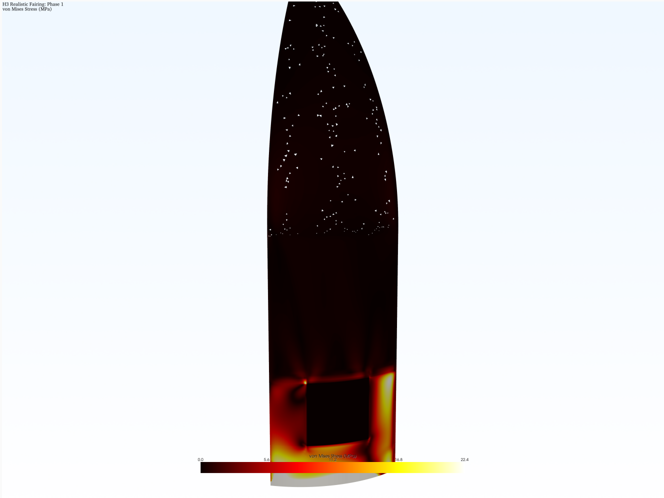
Quantities of interest:

- Peak von Mises stress (limit 1,200 MPa)
- Max cohesive damage SDEG (limit 0.5)
- Max displacement



Non-intrusive: only input files and output databases are touched — no solver source is modified.

Abaqus FE solve — von Mises stress field



- Each PCE quadrature node is one **deterministic Abaqus solve**.
- Real **CFRP / Al-honeycomb** sandwich fairing (Abaqus ODB); von Mises peaks around the **access-door cutout**.
- The PCE study wraps a representative **sandwich panel** of this structure as a black box.

Result 1: response statistics converge at degree 2

QoI	Mean	Std. Dev.	CV	P_f	β
$\sigma_{\mathrm{vM}}^{\max}$ (deg 2)	209.7 MPa	2.375 MPa	1.13%	0	∞
$\sigma_{\mathrm{vM}}^{\max}$ (deg 3)	209.7 MPa	2.378 MPa	1.13%	0	∞
SDEG	0.000	0.000	—	0	∞
$u^{\max}$	10.07 mm	0.178 mm	1.76%	—	—

Degree 2 (66 runs) and degree 3 (286 runs) agree to **four significant figures**.

This is more than a convergence check — it *quantifies the curvature* of the response surface. Had higher-order terms mattered, the extra 220 simulations would have moved the standard deviation. They did not.

Degree 2 is the **correct model order**, not an economy measure.

SDEG $\equiv 0$ in all **352** runs.

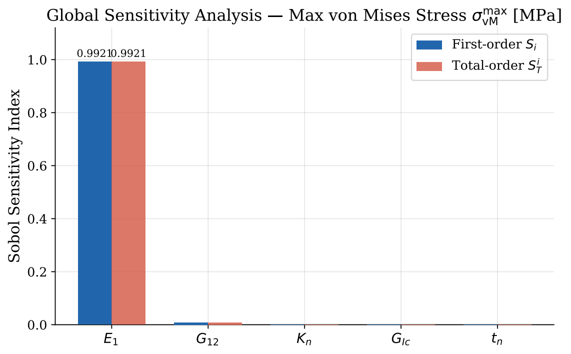
Result 2: E_1 governs over 99% of the variance

	S_i stress	S_i disp.
E_1	**0.9921**	**0.9995**
G_{12}	0.0079	0.0005
K_n, G_{Ic}, t_n	≈ 0	≈ 0

$S_i \approx S_T^i$: inputs act additively, no resolvable interactions.

The 5-D problem collapses to **effectively 1-D** in E_1.

G_{12} carries *twice* the CoV of E_1 yet contributes $< 1\%$ — sensitivity, not scatter, sets the ranking.



Why does E_1 dominate? A first-order argument

$$\mathrm{Var}(Y) \approx \sum_{i=1}^{d} \left(\frac{\partial Y}{\partial \xi_i} \right)^2 \sigma_{\xi_i}^2$$

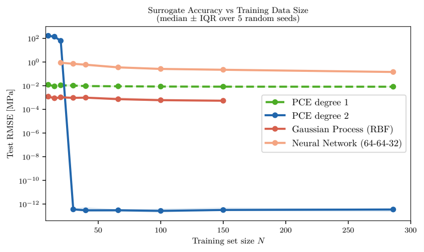
- Each contribution scales with **sensitivity** $\times$ **variance** — not either alone.
- Panel carries load by **membrane action**: stress and deflection set by axial stiffness $E_1 t \Rightarrow$ large $\partial Y / \partial E_1$, and E_1 still carries 5% scatter.
- G_{12} has *twice* the CoV but tiny $\partial Y / \partial G_{12} \Rightarrow$ suppressed.

Ranking is set by which uncertainty the structure is configured to *amplify* — not by which input is most uncertain.

Result 3: PCE vs neural-network surrogate

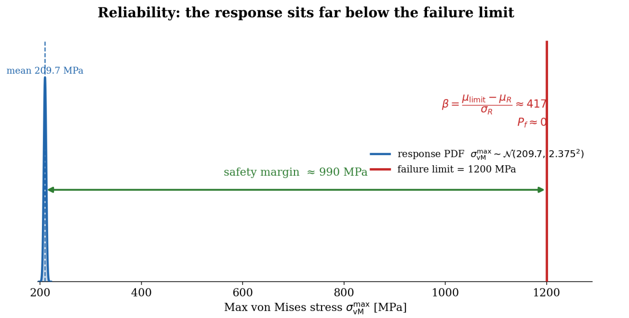
LOO RMSE	PCE deg 2	PCE deg 3	NN (66)	NN (286)
$\sigma_{\mathrm{vM}}^{\max}$ (MPa)	9.4×10^{-4}	1.5×10^{-3}	7.2×10^{-4}	$\mathbf{1.8 \times 10^{-2}}$
$u^{\max}$ (mm)	4.2×10^{-5}	7.1×10^{-5}	2.7×10^{-5}	$\mathbf{1.6 \times 10^{-3}}$

- At $N = 66$ both are accurate to $\sim 10^{-3}$ MPa — the NN marginally lower on this metric.
- The difference is **stability**: NN error **grows** 25× from 66 to 286 points, while PCE deg. 2 holds; deg. 3 is slightly *worse* (over-parameterisation).
- PCE returns Sobol indices analytically; the NN would need a separate Monte Carlo study.



The NN curve turns *upward* — the overfitting signature.

Reliability: a wide margin, and an unidentifiable interface



A degree-2 PCE surrogate places the entire stress distribution far below the limit; cohesive damage never initiates (SDEG $= 0$).

$$P_f = P(\sigma_{\text{vM}}^{\max} > 1200) = 0, \quad \beta = \infty$$

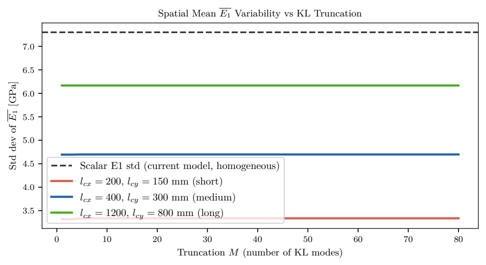
- Mean stress is $(1200 - 209.7)/2.38 \approx \mathbf{416}\sigma$ below failure.
- Read $P_f = 0$ as "**below the resolution** of the analysis" (MC of 10^6; inputs truncated at $\pm 50\%$).
- SDEG $\equiv 0 \Rightarrow$ cohesive parameters are **structurally unidentifiable** at this load.

Extension T9: non-Gaussian KL random field for E_1

KL expansion of an anisotropic exponential kernel, then:

- Gaussian field admits **negative stiffness** $\Rightarrow$ **translation field** $H = F^{-1}(\Phi(Z))$: lognormal, positive by construction.
- Preserves rank, *not* linear correlation: $\rho_H = \dfrac{e^{\sigma_{\ln}^2 \rho_Z} - 1}{e^{\sigma_{\ln}^2} - 1}$ — distortion 3×10^{-4} (CoV 5%) to 3.8×10^{-2} (60%).
- Spatial averaging: $\sigma(\overline{E_1}) \approx 4{,}700 < 7{,}300$ MPa.

Conservative for **average-governed** responses — not for peak stress.

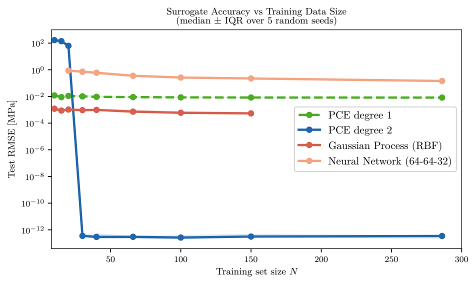


Extension T13: a correct prior beats model flexibility

PCE (deg. 1/2), **GP** (RBF), **NN** (MLP) over N = 10–286. Median RMSE, 5 seeds.

Surrogate	RMSE at N=66
PCE deg. 2	3.0×10^{-13} MPa
GP	7.3×10^{-4} MPa
NN	3.6×10^{-1} MPa

Reference response *is* a deg.-2 polynomial $\Rightarrow 10^{-13}$ measures **exact representability**, not generalisation. The meaningful factor (≈500×) is **GP vs NN**.



GP epistemic std 0.008 MPa vs aleatory 2.37 MPa = **0.3%**.

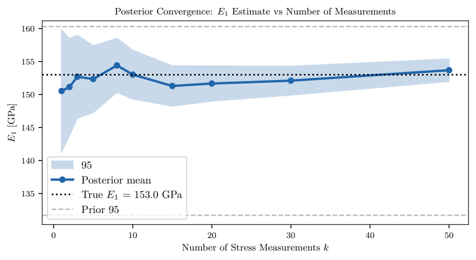
Extension T6: E_1 dominance $\Rightarrow$ tractable Bayesian identification

PCE linear term: $\hat{\sigma}_{\mathrm{vM}} = 209.7 + 2.365\,\xi_{E_1}$ MPa. Conjugate Gaussian update on the mean of k measurements:

$$\sigma_{\mathrm{post}}(E_1) = \frac{\sigma_{E_1}}{\sqrt{1 + k\,c_1^2/\sigma_{\mathrm{meas}}^2}}$$

Key result: posterior std falls from 7,300 MPa to 2,582 (k=5), **1,885** (k=10, −74%), 1,356 (k=20).

Reduces **epistemic** uncertainty $\Rightarrow$ tolerance allocation. P_f is already 0 at 416σ, so reliability is *unchanged*.



Conclusions & limitations

- PCE deg. 2 (66 runs) converged; E_1 **controls** $> 99\%$ of variability; panel reliable at 416σ.
- **T9**: lognormal translation field guarantees positivity; spatial averaging cuts $\sigma(\overline{E_1})$ to 4,700 MPa — conservative for *average-governed* responses.
- **T13**: a correct prior beats flexibility at small N; the 10^{-13} is exact representability, the meaningful gap (500×) is GP vs NN.
- **T6**: 10 measurements $\Rightarrow -74\%$ posterior std — **epistemic** reduction, for tolerance allocation, not for reliability.

Honest limitations

$P_f = 0$ is resolution-limited; inputs assumed independent; single panel, not full fairing; cohesive parameters unidentifiable at this load; KL realisations never propagated through the solver. **Next:** activate cohesive zone; correlated input model; map the field into the FE run.